

ARUNIT BAIDYA

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Master of Science, Computer Science

Aug 2024 – April 2026

Boston, MA, US

Anna University, PSG College Of Technology

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 – July 2022

Coimbatore, India

WORK EXPERIENCE

Arcesium | *Data, Backend, Cloud-Native*

Software Engineer

Jul 2022 - Jul 2024, Bangalore

- Profiled and optimized **Python & Java code** performing **Spark** operations in end-to-end big data system to meet client SLOs (≤ 10 min time to output with 1,000–100,000 records per input file), **improving process times by 40%**.
- Built JSON config-driven data-filtering module for Spark ETL in **reducing compute usage by 30%** via partition pruning.
- Developed **event-driven microservices** with AWS SNS / SQS, optimizing batch size & visibility timeout to **cut latency by 25%**.
- Optimized resource provisioning sizes (T-shirt model) in **ArgoWorkflow** and **Kubernetes**, **adopted in 10+ workflows**.
- Automated CI/CD pipelines with **Jenkins, Docker, & Kubernetes**, integrating JUnit for *testing ensuring 85% coverage*.
- Implemented **SQL** for SCD type 2 merge, enabling bi-temporality & historical querying, state of data at point in time.
- Created LLD documentation with business logic translation, **reducing** cross-functional team **inquiries by 50%**.

Software Engineering Intern

Feb 2022 - May 2022, Remote

- **Increased ETL throughput by 70%** via preprocessing stage to split heterogeneous data for parallel processing.
- Boosted ETL reliability by engineering module in **Python** to extract malformed data to global kickouts Delta-table.
- Enhanced system observability by integrating log publishing to **Elasticsearch** and creating **Datadog** monitors.

PROJECTS

Scalable Image Sharing Platform (GitHub) | *Golang, JavaScript, Kafka, Redis, Postgres, MinIO/S3*

- Architected a high-performance Golang backend utilizing **PostgreSQL** and **MinIO/S3** for scalable, reliable storage.
- Developed **RESTful APIs** for image uploads, likes, dislikes, views, and comments with **batched database updates**.
- Leveraged **Kafka** for event-driven processing and implemented **Redis** caching (TTL & LRU eviction) to minimize latency.
- Deployed using **Docker**, with **JWT authentication, rate limiting, & Prometheus** monitoring for enhanced observability.

Image Processing Application (GitHub) | *Java, SOLID, MVC*

- Developed an image processing application in **Java** supporting CLI, script, and GUI inputs, demonstrating **SOLID principles**, higher order functions, **object-oriented design, MVC**, design patterns, and **test-driven development**.

Scalable TODO-List API (GitHub) | *Golang, Kafka, Redis, Docker, Ansible*

- Designed a scalable REST API for a TODO-List app with **PostgreSQL**, leveraging Redis caching, paginated list queries, event-driven architecture, and automated deployments via **Ansible** and **GitHub CI/CD**.

egOS (GitHub) | *C, Operating Systems, RISC V, Multithreaded Programming*

- Enhanced a RISC-V-based operating system (egOS) by implementing **user-level threading**, a **multilevel feedback queue** (MLFQ) kernel scheduler, **physical memory protection**, exception handling, and **virtual memory** management via paging in **C**.

Simple Linux Shell (GitHub) | *C, Linux*

- Built a Linux shell in **C** with command execution for internal & external commands, piping, & I/O redirection using **fork**.

SKILLS

Programming Languages:	Python, Java, Go, C, JavaScript, HTML, CSS
Frameworks and Libraries:	REST, Redis, Swing, Pytest, JUnit, Argo Workflow, Ansible, Prometheus
Data Engineering:	Spark, Kafka, ETL, DeltaLake, Snowflake, PostgresDB, MySQL, NoSQL, RDBMS
Cloud and DevOps:	REST API development, AWS, Elasticsearch, SNS, SQS, Terraform, Docker, Kubernetes
Software Development:	CI/CD, Agile Development, Test-Driven Development, Code Reviews, Git